

Does Partisan Attachment Amplify Affective Polarization?*

Vertical and Horizontal Dimensions Across 25 European Democracies

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Affective polarization (AP) research faces a persistent measurement challenge: should partisans be identified through self-reported attachment or vote choice? This distinction matters theoretically – identity accounts predict attachment amplifies polarization, while categorization accounts suggest vote anchors suffice – and comparatively, as explicit attachment rates vary widely across European democracies. We introduce *partisan type* (explicit versus implicit partisans) and test whether subjective attachment differentially predicts *vertical* AP (party-targeted attitudes) versus *horizontal* AP (citizen-targeted behavior). Analyzing thermometer ratings and conjoint trust-game data from 25 countries (N 29,800), we find explicit partisans exhibit significantly higher vertical AP than implicit partisans, but the two groups show no difference in horizontal behavioral discrimination. These results reveal distinct psychological mechanisms: subjective attachment shapes party affect but not interpersonal behavior. For comparative research, our findings validate combining vote-anchored and attachment-based samples for behavioral studies while cautioning that restricting to explicit identifiers may misrepresent attitudinal polarization’s behavioral translation in multiparty systems.

1 Introduction

Affective polarization (AP) – the tendency to favor one’s own partisan group while disliking opponents – has emerged as a defining feature of contemporary democracies (Iyengar, Sood, and Lelkes 2012; Iyengar et al. 2019; Boxell, Gentzkow, and Shapiro 2024). Beyond policy disagreement, AP encompasses social distancing, interpersonal hostility, and discriminatory

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behavior toward partisan outgroups, raising concerns about democratic cooperation and political stability (Graham and Svulik 2020; Kingzette et al. 2021; Lars Eric Berntzen, Kelsall, and Harteveld 2023). Extensive research now documents AP’s prevalence across advanced democracies (Wagner 2021; Reiljan 2020; Reiljan et al. 2024), yet fundamental questions persist about its measurement and psychological foundations. A particularly consequential ambiguity concerns who counts as “partisan” in empirical analyses: should researchers restrict samples to citizens who self-report feeling attached to a party, or include all those who reveal partisan preferences through vote choice regardless of reported attachment? This operationalization choice carries major implications for both theory and comparative inference, yet existing research has not systematically examined whether these two groups exhibit equivalent levels of AP.

The partisan identification question matters theoretically because it implicates competing accounts of AP’s underlying mechanisms. Social identity perspectives emphasize subjective attachment as a psychological bond that intensifies emotional investment in political groups, predicting that explicit partisan identifiers should exhibit stronger polarization than those who merely vote without reporting felt attachment (Leonie Huddy, Mason, and Aarøe 2015; Mason and Wronski 2018; Iyengar and Westwood 2015). In contrast, categorization and cue-processing accounts suggest that partisan labels and vote anchors provide sufficient information to generate polarized responses even absent subjective identification (Orr, Fowler, and Huber 2023; Orr and Huber 2020; Dafoe, Zhang, and Caughey 2018). Distinguishing these mechanisms has proven difficult because party cues simultaneously signal group membership and ideological content, producing observational equivalence (Orr, Fowler, and Huber 2023). The partisan type distinction—explicit versus implicit partisanship—provides analytical leverage: systematic differences between these groups would support identity-centric accounts, while equivalent behavior would challenge

the assumption that subjective attachment drives polarization.

The measurement stakes are especially high in comparative multiparty research. Unlike the U.S. two-party system, European electorates contain substantial proportions of voters who express stable partisan preferences yet deny feeling attached to any party. The prevalence of explicit partisan attachment varies dramatically—ranging from roughly 40% to 80% across European countries in our data—meaning that restricting analyses to self-identified partisans reshapes the population of inference differently in each national context and undermines cross-national comparability (Röllicke 2023; Mariano Torcal and Comellas 2025). Moreover, widely used AP measures diverge precisely in their treatment of partisans: Reiljan (2020)’s Affective Polarization Index (API) focuses on reported identifiers, while Wagner (2021)’s spread measure encompasses all respondents, yielding potentially different empirical portraits of polarization (Campos and Federico 2025; Carlin and Love 2025). Without systematic evidence on whether explicit and implicit partisans exhibit equivalent AP, researchers face arbitrary decisions about sample inclusion that may distort substantive conclusions and comparative inference.

Recent theoretical refinements further complicate these issues by distinguishing *vertical* AP—attitudes toward parties and elites—from *horizontal* AP—behavior toward fellow citizens (Klar, Krupnikov, and Ryan 2018; Areal and Harteveld 2024; J. Druckman and Levendusky 2019). This differentiation matters because different manifestations of polarization may arise from different psychological mechanisms and carry distinct normative implications (Lars Erik Berntzen 2025; Campos and Federico 2025). If subjective partisan attachment shapes how citizens feel about *parties* but not how they treat partisan *others* in interpersonal contexts, then vertical and horizontal AP would require separate theoretical accounts. Testing whether partisan type differentially predicts these two dimensions could reveal whether attachment-driven and categorization-driven processes operate in

parallel, with the former dominating party-targeted attitudes and the latter governing citizen-targeted behavior—a possibility existing research has not explored.

We address these challenges by distinguishing *explicit* partisans—respondents who report feeling attached to a party—from *implicit* partisans—those who deny such attachment yet reveal partisan anchors through vote choice. We term this distinction *partisan type*. This conceptualization extends classic insights on independents, leaners, and implicit partisan identity (Petrocik 2009; Kinder and Kalmoe 2017; Theodoridis 2017; Elliott 2024) to multiparty systems, isolating a theoretical quantity prior work has not directly examined: whether self-reported attachment enhances polarization beyond the effects of partisan categorization alone. Empirically, we analyze thermometer ratings (vertical AP) and conjoint trust-game experiments (horizontal AP) from 25 European democracies (N 29,800), applying Bayesian hierarchical models with country-level random effects to estimate AP conditional on partisan type while accounting for respondent clustering and cross-national heterogeneity.

Our (preliminary and exploratory) findings reveal a striking asymmetry. For vertical AP measured through party thermometer ratings, explicit partisans exhibit systematically and substantially higher polarization than implicit partisans across all 25 countries. Yet for horizontal AP measured through token allocations in trust games embedding randomized partisan cues, explicit and implicit partisans discriminate identically: both groups show equivalent levels of AP. This vertical-horizontal divergence challenges theories positing unified mechanisms underlying affective polarization. Subjective attachment shapes *attitudes toward parties* but does not amplify *behavioral discrimination among citizens*.

Extending these findings, we plan to examine individual-level moderators to identify what drives vertical AP and whether the explicit–implicit gap varies systematically with respondent characteristics. In our future analyses, we will incorporate theoretically relevant

covariates – including ideological extremity, political interest, social identity strength, perceived threat, democratic norms, social trust, and media consumption – that are available in our data, following insights from existing literature. This approach will allow us to explore whether differences in vertical AP between explicit and implicit partisans are amplified or attenuated among particular subgroups.

These exploratory findings carry several preliminary implications for AP research. First, regarding measurement and population definition, the equivalence of horizontal AP between explicit and implicit partisans suggests that combining vote-anchored and attachment-based respondents is reasonable when estimating behavioral polarization, avoiding unnecessary reductions in external validity. Second, the vertical–horizontal asymmetry indicates that subjective attachment amplifies party-targeted attitudes but does not appear to drive interpersonal discrimination, highlighting the need for multidimensional frameworks distinguishing elite- versus citizen-targeted polarization. Finally, the consistency of this asymmetry across 25 European democracies suggests that these patterns may reflect structural features of multiparty electoral systems rather than country-specific idiosyncrasies, providing a foundation for future analyses of individual-level moderators.

The paper proceeds as follows. Section 2 reviews theoretical perspectives on partisan identity and affective polarization, develops our conceptualization of partisan type, and derives testable hypotheses distinguishing vertical from horizontal AP. Section 3 describes the data from 25 European democracies, the measurement strategy applying Reiljan (2020)’s API logic consistently across both dimensions, and the Bayesian hierarchical modeling approach. Section 4 outlines the planned presentation of empirical results, Section 5 presents the mentioned preliminary descriptive findings.

2 Theory

2.1 Origin and Expansion of Affective Polarization Research

The contemporary study of political polarization was reshaped by the insight that partisan conflict is driven not only by ideological disagreement but also by social and emotional divisions (Iyengar, Sood, and Lelkes 2012). Early work demonstrated that hostility between partisans intensified even while policy preferences remained comparatively stable (Fiorina, Abrams, and Pope 2006; Fiorina, Abrams, and Pope 2008), redirecting scholarly attention from ideological distance (Poole and Rosenthal 1984; McCarty, Poole, and Rosenthal 2006) toward affective orientations between political groups. Building on social identity approaches, affective polarization (AP) is typically defined as the tendency of individuals to feel warmth toward perceived ingroup members and hostility toward outgroups (Iyengar and Westwood 2015; Iyengar et al. 2019; Iyengar and Wagner 2025).

Rooted in Social Identity Theory and Self-Categorization Theory (H. Tajfel and Turner 1979; Henri Tajfel and Turner 2004), AP reflects a broader and well-documented human propensity for ingroup favoritism and outgroup derogation observable even in minimal group settings (Henri Tajfel et al. 1971; Sherif et al. 1961; Brewer 1999). Political science research focuses on cases where the salient social identity is partisanship (L. Huddy 2001; Leonie Huddy, Mason, and Aarøe 2015; Mason 2015). Over the past decade, this perspective has generated a large comparative literature documenting the prevalence of AP across advanced democracies (Wagner 2021; Reiljan 2020; Reiljan et al. 2024; Boxell, Gentzkow, and Shapiro 2024), identifying its drivers (Hobolt, Lawall, and Tilley 2024; Bail et al. 2018; Orr and Huber 2020; Mason 2015; Törnberg 2022; Luttig 2017), tracing its consequences for democratic attitudes and behavior (Diermeier and Li 2019; Lars Eric Berntzen, Kelsall, and Hartevelt 2023; Kingzette et al. 2021; Graham and Svulik 2020;

McConnell et al. 2018; Broockman, Kalla, and Westwood 2023), examining temporal trends (Iyengar and Krupenkin 2018), and evaluating interventions designed to reduce partisan hostility (Adams et al. 2023; Kalla and Broockman 2022; J. N. Druckman, Green, and Iyengar 2023; Mernyk et al. 2022).

The rapid growth of this research program has produced substantial empirical knowledge but also conceptual and measurement disputes (Campos and Federico 2025; Bakker and Lelkes 2024; Iyengar and Wagner 2025; Carlin and Love 2025). Scholars increasingly recognize that the concept encompasses multiple related phenomena and that different operationalizations can yield divergent conclusions about its magnitude, distribution, and implications. Consequently, cumulative knowledge now depends less on documenting whether AP exists than on clarifying what precisely is being measured and for whom theoretical claims apply.

2.2 Core Debates and Conceptual Refinements

Early debates centered on whether polarization should be understood as a state or a process and whether its democratic consequences are uniformly harmful or conditional (Kingzette et al. 2021; Moore-Berg et al. 2020; Westwood et al. 2022). More recent work emphasizes that different manifestations of AP may have distinct normative implications and psychological origins (Lars Erik Berntzen 2025; Broockman, Kalla, and Westwood 2023; Voelkel et al. 2023). Three major refinements structure the contemporary literature.

First, target differentiation. Scholars distinguish vertical polarization—affect toward parties and elites—from horizontal polarization—affect and behavior toward fellow citizens (Klar, Krupnikov, and Ryan 2018; Harteveld 2021; J. Druckman and Levendusky 2019; Areal and Harteveld 2024). Vertical AP can further target parties or leaders (Reiljan

et al. 2024; M. Torcal and Comellas 2022), while horizontal AP concerns interpersonal discrimination. Because elite-directed hostility and citizen-directed behavior may arise from different mechanisms, conflating them risks obscuring causal relationships.

Second, manifestation differentiation. Research differentiates attitudinal AP, typically measured with feeling thermometers or social distance scales, from behavioral AP, measured through economic allocation decisions or real interactions (J. Druckman and Levendusky 2019; Carlin and Love 2025; Iyengar and Westwood 2015). These measures need not align: citizens may express dislike without discriminating behaviorally, or discriminate without reporting strong hostility.

Third, operationalization choices. Measurement depends on defining partisan groups. Scholars must determine how to identify partisans (attachment vs vote choice), whether individuals may hold multiple ingroups (Kekkonen and Ylä-Anttila 2021; Kekkonen et al. 2022; Bantel 2023), how to define outgroups (least-liked party vs average opponent) (Wagner 2021), and whether to include non-partisans in analysis (Röllicke 2023). These decisions reshape empirical findings rather than merely refining them.

Together, these developments imply that AP is not a single phenomenon but a family of related outcomes whose interpretation depends heavily on conceptual and measurement choices (Campos and Federico 2025; Iyengar and Wagner 2025).

2.3 Extending Affective Polarization to Multiparty Systems

These challenges intensify in multiparty democracies (Röllicke 2023; Mariano Torcal and Comellas 2025). In two-party systems, ingroup and outgroup boundaries are clear. In fragmented party systems, however, several ambiguities emerge.

First, it is unclear what constitutes the ingroup. Researchers variously define it as the most-liked party, multiple parties, or ideological blocs (Kekkonen and Ylä-Anttila 2021; Kekkonen et al. 2022; Bantel 2023; Reiljan and Ryan 2021). Second, identifying membership can rely on subjective attachment or vote choice (Wagner 2021; Reiljan 2020). Third, defining outgroups may involve the least-liked party or the average evaluation of all opponents (Wagner 2021). Finally, many citizens lack explicit partisan identification, raising the question of whether they belong in the population of inference (Garzia, Ferreira da Silva, and Maye 2023).

Existing measurement strategies address these issues differently. Some approaches assign each respondent a single ingroup and calculate polarization relative to all outparties (Reiljan 2020; Hahm, Hilpert, and König 2024); others measure dispersion in party evaluations and include non-identifiers (Wagner 2021); still others allow multiple ingroups (Kekkonen and Ylä-Anttila 2021; Kekkonen et al. 2022). Because explicit partisan attachment varies substantially across European countries, these decisions affect cross-national comparability. Restricting analyses to self-identified partisans can approximate the electorate in countries with high identification rates but represent only a minority in others. As a result, identical operationalizations may imply different target populations across cases.

Moreover, attachment questions themselves may capture different psychological phenomena depending on party system institutionalization and stability (Garzia, Ferreira da Silva, and Maye 2023). Comparative research typically assumes equivalence, yet this assumption is rarely evaluated.

2.4 Partisan Type, Population of Inference, and Research Contribution

To address these comparability and interpretation problems, we distinguish between respondents who report attachment to a party and those who deny attachment but reveal a partisan anchor through vote choice. We refer to these categories as partisan types: explicit and implicit partisans. The distinction serves primarily as a consistent terminology allowing comparable analysis across contexts rather than a novel theoretical construct.

The distinction matters because it directly affects the population of inference. If the proportion of explicit and implicit identifiers varies across countries, analyses limited to explicit identifiers describe fundamentally different electorates across cases. Consequently, comparative conclusions about AP may depend as much on sampling decisions as on political behavior.

Beyond measurement, the comparison offers leverage for adjudicating competing theoretical accounts of polarization. Social identity theories emphasize subjective attachment as a psychological bond that intensifies emotional investment in political groups ([Leonie Huddy, Mason, and Aarøe 2015](#); [Mason and Wronski 2018](#); [Leonie Huddy 2013](#); [West and Iyengar 2022](#)). In contrast, categorization and cue-processing perspectives suggest that partisan labels and vote anchors alone provide sufficient information to generate polarized responses, even absent felt attachment ([Orr, Fowler, and Huber 2023](#); [Orr and Huber 2020](#); [Dafoe, Zhang, and Caughey 2018](#); [Henri Tajfel et al. 1971](#)).

We therefore compare explicit and implicit partisans across two dimensions of AP: (1) vertical attitudinal polarization toward parties and (2) horizontal behavioral discrimination toward citizens. We further examine whether individual characteristics moderate these relationships. The analysis of individual-level drivers remains exploratory; covariate selection will be guided by existing research on ideological extremity ([Abramowitz and](#)

Saunders 2006, 2008), engagement (Luttig 2017), social identity strength (Mason 2015; Mason and Wronski 2018), perceived threat (Leonie Huddy, Mason, and Aarøe 2015), democratic norms (Graham and Svolik 2020), social trust, and media consumption (Bail et al. 2018; Törnberg 2022).

2.5 Hypotheses

We evaluate competing explanations of AP by comparing explicit and implicit partisans across vertical (attitudinal) and horizontal (behavioral) domains. Social identity accounts emphasize subjective attachment as a psychological bond that intensifies affective investment in political groups (Leonie Huddy, Mason, and Aarøe 2015; Mason and Wronski 2018), whereas categorization accounts argue that partisan labels and vote anchors alone suffice to generate polarized responses (Orr, Fowler, and Huber 2023; Dafoe, Zhang, and Caughey 2018).

2.5.1 Vertical (Attitudinal) AP

If attachment creates emotional investment in political objects, explicit identifiers should evaluate ingroup and outgroup parties more differently than implicit identifiers.

H_{1a} (Identity mechanism): Explicit partisans exhibit higher vertical AP than implicit partisans.

If partisan categorization alone activates evaluative schemas, attachment adds little beyond knowing one’s partisan anchor (Orr, Fowler, and Huber 2023).

H_{1b} (Categorization mechanism): Explicit and implicit partisans exhibit equivalent vertical AP.

2.5.2 Horizontal (Behavioral) AP

If identity strength drives discriminatory behavior toward political opponents, explicit identifiers should favor copartisans more strongly in interpersonal decisions ([Leonie Huddy, Mason, and Aarøe 2015](#); [Iyengar and Westwood 2015](#)).

H_{2a} (Identity mechanism): Explicit partisans exhibit higher horizontal AP than implicit partisans.

If categorization and inferred alignment are sufficient to generate intergroup bias ([Henri Tajfel et al. 1971](#); [Sherif et al. 1961](#)), behavioral discrimination should not depend on subjective attachment.

H_{2b} (Categorization mechanism): Explicit and implicit partisans exhibit equivalent horizontal AP.

2.5.3 Individual-Level Moderation

Identity-based mechanisms imply that affective investment intensifies among individuals whose political orientations are psychologically consolidated ([Mason 2015](#); [West and Iyengar 2022](#)).

H₃: The explicit–implicit gap in vertical polarization is larger among ideologically extreme and politically engaged respondents.

If behavioral discrimination primarily follows categorization rather than identity strength, these moderators should not alter behavioral differences.

H₄: Individual-level moderators affect vertical but not horizontal differences between partisan types.

3 Research Design

3.1 Data

We analyze survey data from 25 European democracies collected as part of a large-scale conjoint experiment fielded in 2019 (N = 29,800 respondents) ([Hahm, Hilpert, and König 2024](#)). The study was designed to examine partisan discrimination across diverse party systems, combining standard survey measures of political attitudes with behavioral games embedding randomized partisan cues. The cross-national scope – spanning established Western European democracies, Southern European countries, and newer Central and Eastern European democracies – provides substantial variation in party system characteristics, democratic consolidation, and patterns of partisan attachment.

Each respondent completed: (1) standard demographic and political attitude batteries, including party thermometer ratings and a partisan attachment item; (2) vote choice questions; and (3) behavioral games (dictator and trust games) in which they allocated resources to profiles randomly assigned partisan and demographic attributes. The hierarchical structure of the data – with multiple evaluations nested within respondents, who are nested within countries – requires multilevel modeling to appropriately account for clustering and estimate cross-national heterogeneity.

3.2 Measuring Vertical Affective Polarization: Party Thermometer

Ratings

Vertical (attitudinal) AP is measured using feeling thermometer ratings toward political parties. Respondents evaluated all major parties in their country on a 0 (dislike) to 100 (like) scale. We rescale this measure to a 0–10 scale so that the outcome space matches

the horizontal AP token allocation measure, facilitating direct comparison across the two dimensions.

The hierarchical structure generates multiple observations per respondent: each individual i in country c provides ratings r for multiple parties. We construct a binary indicator CoPartisan_{ric} that equals 1 when the rated party matches respondent i 's partisan anchor and 0 otherwise. This operationalization allows us to estimate AP as the difference between ingroup affect (ratings of one's own party) and average outgroup affect (ratings of all other parties), directly implementing Reiljan (2020)'s AP Index (API) logic within a regression framework.

3.3 Measuring Horizontal Affective Polarization: Token Allocation in Behavioral Games

Horizontal (behavioral) AP is measured through token allocations in conjoint experiments embedding partisan cues (Hahm, Hilpert, and König 2024). Respondents participated in dictator and trust games where they allocated tokens to profiles describing hypothetical citizens. Each profile included randomized attributes: partisan affiliation, age, social class, gender, religion, and EU position. The randomization ensures that partisan cues are orthogonal to other profile characteristics, isolating the causal effect of copartisanship on resource allocation.

In the dictator game, respondents divided 10 tokens between themselves and a partner. In the trust game, respondents decided how many tokens (0–10) to send to a partner, with sent tokens tripled and the partner choosing how many to return. Both games measure behavioral discrimination: the extent to which respondents favor copartisans over outpartisans in resource allocation decisions.

As with thermometer ratings, the design generates multiple observations per respondent: each individual i evaluated multiple profiles r . We construct $\text{CoPartisan}_{ric} = 1$ when the profile’s party affiliation matches respondent i ’s partisan anchor, and 0 otherwise. This parallel operationalization allows direct comparison of vertical and horizontal AP using identical analytical frameworks.

3.4 Partisan Type: Explicit Versus Implicit Partisans

Our key explanatory variable distinguishes respondents by partisan type. The survey first asked: “Do you consider yourself to be close to any particular political party? If so, which party do you feel close to?” Respondents indicating closeness to any party are classified as explicit partisans ($\text{Type}_i = 1$). For these individuals, the named party defines the respondent’s partisan anchor. This group comprises roughly 18,000 respondents in our sample.

Respondents who selected the option indicating no attachment to any party were subsequently evaluated using multiple electoral preference items capturing both retrospective and prospective behavior: vote intention and reported vote choice (including alternative and earlier election recall items). We employ a recall-maximizing strategy: if a respondent reported any party preference across these measures, that party is assigned as the respondent’s partisan anchor and the respondent is classified as an implicit partisan ($\text{Type}_i = 0$). These individuals deny subjective identification but reveal a behavioral partisan anchor through electoral preference (approximately 6,000 respondents).

Respondents who neither reported party attachment nor indicated any party in any vote item are classified as non-partisans (approximately 3,900 respondents). Because they lack a defensible partisan anchor, they cannot be assigned copartisan–outpartisan

relationships and are therefore excluded from analyses requiring such comparisons.

For explicit and implicit partisans alike, the identified party—whether derived from attachment or electoral preference—serves as the partisan anchor defining copartisan versus outpartisan distinctions in subsequent evaluations. This procedure distinguishes subjective identification from behavioral anchoring while maintaining a consistent reference point for measuring AP across respondents.

Importantly, the distribution of partisan types varies substantially across countries. Figure 1 visualizes this cross-national heterogeneity in partisan attachment and behavioral anchoring. The share of explicit partisans ranges from approximately 85% of respondents in Denmark to about 41% in Latvia, while implicit partisans show the inverse pattern, varying from roughly 41% in Latvia to 8% in Denmark. Non-partisans also differ markedly across contexts, ranging from 23% in Croatia to 6% in Estonia.

These patterns indicate that both reported attachment and overall partisan composition differ strongly across national contexts. Consequently, identical sampling decisions – such as restricting analyses to self-reported identifiers – would generate different populations of inference across countries. This cross-national variation motivates our comparison of AP across partisan types rather than focusing exclusively on explicit identifiers.

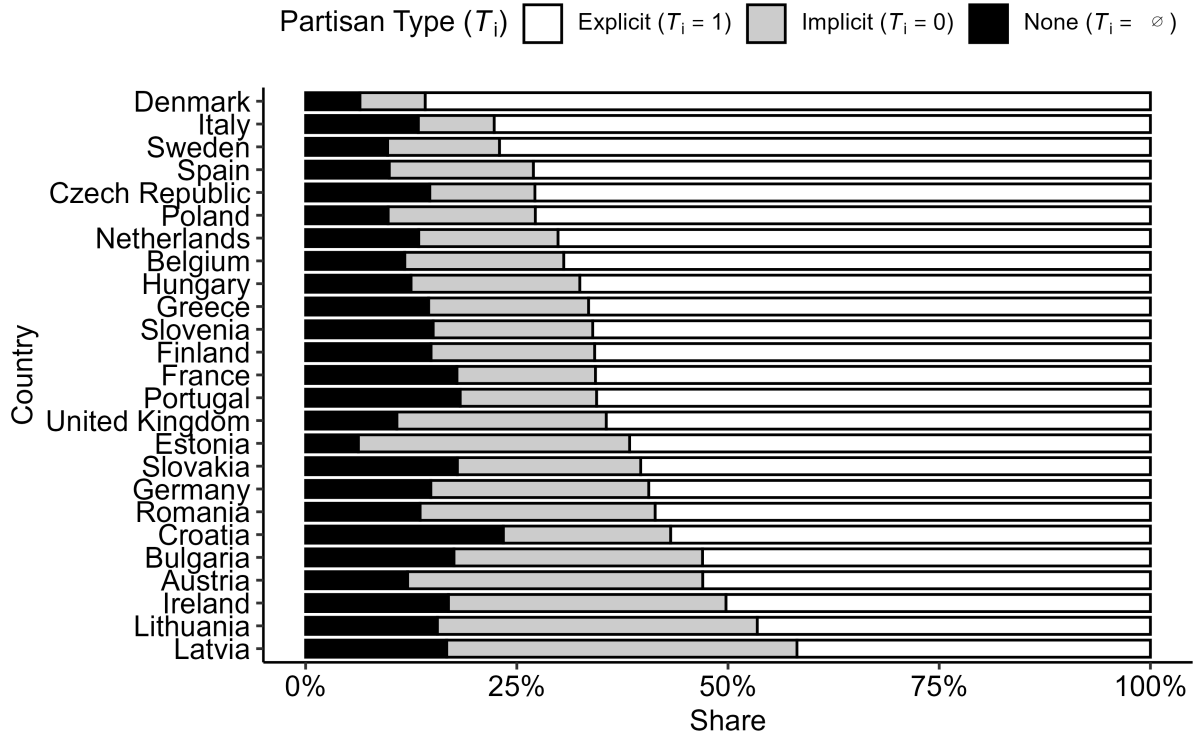


Figure 1: Distribution of partisan types, by country. Stacked horizontal bars show the within-country share (%) of three partisan types: explicit partisans (respondents who reported a subjective attachment to a party, $T_i = 1$), implicit partisans (respondents who reported no attachment but did report a vote preference or intention, $T_i = 0$), and respondents who reported neither (none, $T_i = \emptyset$). Percentages sum to 100% within each country, with country samples containing about 1,100 respondents each (detailed numbers are reported in appendix section X).

Source: [Code Notebook 3.1](#)

3.5 Analytical Strategy

We (plan to) employ Bayesian hierarchical models to estimate AP as a function of partisan type, progressively building complexity to balance interpretability with inferential rigor. Our approach unfolds in three stages: first, we introduce the core linear predictor that defines AP and connects directly to our hypotheses. Second, we embed this predictor within a multilevel structure enabling cross-national inference. Third, we extend the model to examine individual-level moderators of partisan type effects. This progression

maintains a constant estimand – the ingroup-outgroup evaluation gap – while progressively incorporating the nested data structure and testing conditional relationships.

3.5.1 The Linear Predictor: Defining Affective Polarization

At its core, our analysis asks whether respondents evaluate copartisans differently from outpartisans, and whether this difference differs among partisan types. The basic model specification is:

$$\text{Outcome}_{ric} = \alpha + \beta_1 \cdot \text{CoPartisan}_{ric} + \beta_2 \cdot \text{Type}_i + \beta_3 \cdot (\text{CoPartisan}_{ric} \times \text{Type}_i)$$

where r indexes evaluations (party ratings or behavioral allocations), i indexes respondents, and c indexes countries. The outcome is either a feeling thermometer rating (vertical AP) or token allocation in the behavioral task (horizontal AP). The indicator $\text{CoPartisan}_{ric} = 1$ when the evaluated party or profile matches respondent i 's partisan anchor, and $\text{Type}_i = 1$ for explicit partisans versus 0 for implicit partisans.

This specification defines AP as a derived quantity of interest: the expected difference in evaluations between copartisans and outpartisans. Formally:

$$\text{AP}_t = \mathbb{E}[\text{Outcome} \mid \text{CoPartisan} = 1, \text{Type} = t] - \mathbb{E}[\text{Outcome} \mid \text{CoPartisan} = 0, \text{Type} = t]$$

From the linear predictor, this translates to $\text{AP}_{\text{implicit}} = \beta_1$ for implicit partisans and $\text{AP}_{\text{explicit}} = \beta_1 + \beta_3$ for explicit partisans. The central hypothesis test therefore concerns

β_3 : identity-based theories predict $\beta_3 > 0$, indicating that explicit attachment amplifies polarization, while categorization-based theories predict $\beta_3 \approx 0$, indicating that partisan anchors alone suffice to generate the ingroup-outgroup gap. We apply this framework separately to vertical and horizontal outcomes, enabling direct comparison of partisan type effects across attitudinal and behavioral polarization.

This operationalization maintains consistency with Reiljan (2020)’s AP Index – calculating polarization as ingroup affect minus average outgroup affect – while embedding the comparison within a regression framework that isolates the role of subjective attachment. The binary copartisan indicator ensures that AP reflects the difference between a single ingroup evaluation and the average across all outgroup evaluations, exactly paralleling the API logic but extending it to both thermometer ratings and behavioral allocations.

3.5.2 Multilevel Structure for Cross-National Inference

Because evaluations are nested within respondents and respondents within countries, we estimate the linear predictor within a three-level hierarchical model:

$$\begin{aligned} \text{Outcome}_{ric} = & (\alpha + \alpha_i + \alpha_c) \\ & + (\beta_1 + \beta_{1c}) \cdot \text{CoPartisan}_{ric} \\ & + (\beta_2 + \beta_{2c}) \cdot \text{Type}_i \\ & + (\beta_3 + \beta_{3c}) \cdot (\text{CoPartisan}_{ric} \times \text{Type}_i) \\ & + \varepsilon_{ric} \end{aligned}$$

where $\alpha_i \sim \mathcal{N}(0, \sigma_{\text{respondent}}^2)$ represents respondent-specific intercepts accounting for repeated measures within individuals, $\alpha_c \sim \mathcal{N}(0, \sigma_{\text{country}}^2)$ represents country-specific intercepts, and $\beta_{1c}, \beta_{2c}, \beta_{3c}$ represent country-varying slopes for the copartisan effect, type

effect, and their interaction. Residual variation $\varepsilon_{ric} \sim \mathcal{N}(0, \sigma^2)$ captures within-respondent noise across evaluations.

This hierarchical specification serves two purposes. First, it appropriately accounts for clustering: standard errors reflect uncertainty at all levels of the data structure, avoiding anti-conservative inference. Second, it enables estimation of both population-average and country-specific effects. Population-average AP is recovered by marginalizing over random effects, yielding $AP_t = \beta_1 + t \cdot \beta_3$, which answers whether explicit partisans polarize more than implicit partisans on average across European democracies. Country-specific AP is estimated by conditioning on country-level random effects, yielding $AP_{c,t} = (\beta_1 + \beta_{1c}) + t \cdot (\beta_3 + \beta_{3c})$, which reveals whether the explicit-implicit gap varies systematically across party systems.

The multilevel approach is particularly important given the substantial cross-national variation in explicit partisan prevalence documented in Figure 1. If partisan type effects vary systematically with party system characteristics – for instance, if subjective attachment matters more in institutionalized party systems – the country-varying slopes will capture this heterogeneity. Conversely, if the vertical-horizontal asymmetry in partisan type effects holds consistently across all 25 countries, this would suggest a broadly shared mechanism operating across diverse institutional contexts. We assess this through posterior distributions of β_{3c} , testing whether country-specific deviations cluster around zero or exhibit systematic patterns.

3.5.3 Individual-Level Moderators of Partisan Type Effects

To examine whether the explicit–implicit gap in vertical AP varies with respondent characteristics, we extend the linear predictor to include individual-level moderators and

their interactions with partisan type:

$$\begin{aligned}
\eta_{ric} = & \alpha + \beta_1 \cdot \text{CoPartisan}_{ric} + \beta_2 \cdot \text{Type}_i + \beta_3 \cdot \text{Covariate}_i \\
& + \beta_4 \cdot (\text{CoPartisan}_{ric} \times \text{Type}_i) \\
& + \beta_5 \cdot (\text{CoPartisan}_{ric} \times \text{Covariate}_i) \\
& + \beta_6 \cdot (\text{Type}_i \times \text{Covariate}_i) \\
& + \beta_7 \cdot (\text{CoPartisan}_{ric} \times \text{Type}_i \times \text{Covariate}_i)
\end{aligned}$$

The key parameter is β_7 , which evaluates whether the partisan-type difference in AP (β_4) varies systematically across individuals. Substantively, this parameter indicates whether the explicit–implicit gap depends on respondent characteristics rather than representing a uniform population-level difference.

For any given value of a covariate, the partisan-type gap in AP is

$$\Delta\text{AP} = \beta_4 + \beta_7 \cdot \text{Covariate}.$$

We will therefore interpret moderation using simulated conditional marginal effects derived from the posterior distribution, tracing how the explicit–implicit difference changes across the observed covariate range and identifying regions where the gap is credibly positive, negative, or indistinguishable from zero.

At the present stage, the specific covariates to be included are not fixed. Their selection will be guided by prior research on correlates of AP and by data availability across countries. Models will be estimated separately for vertical (thermometer) and horizontal (behavioral allocation) outcomes to evaluate whether moderation appears in attitudinal polarization but not behavioral discrimination.

The moderator specification can be combined with the multilevel structure introduced above, allowing effects to vary across countries. For clarity, we begin with pooled moderator models that estimate average patterns across contexts; country-varying moderator effects constitute a planned extension examined in supplementary analyses.

3.5.4 Summary

Our three-stage analytical strategy maintains a constant estimand – AP as the ingroup-outgroup evaluation gap—while progressively improving inferential realism. Stage 1 defines AP through a simple interaction-based linear predictor that connects directly to competing theoretical predictions about partisan type effects. Stage 2 embeds this predictor within a hierarchical model accounting for data structure and enabling cross-national inference about heterogeneity in partisan type effects. Stage 3 extends the framework to test whether individual characteristics moderate the explicit-implicit gap, distinguishing contexts where subjective attachment matters most. This progression balances transparency, interpretability, and statistical rigor, allowing us to adjudicate between identity-based and categorization-based accounts of AP across both vertical and horizontal dimensions.

4 Planned Results Structure

4.1 Vertical AP

- Figure/Panel with pooled EU25-wide and country specific AP by type

4.2 Horizontal AP

- Figure/Panel with pooled EU25-wide and country specific AP by type

4.3 Individual level moderation

- Marginal/Conditional effect figure(s), with covariate values on x-axis, effect on AP on y axis, different lines (and intervals) by type, separate panels for vertical/horizontal AP
- figure with panels by country? Unsure yet

5 Exploratory descriptive results

5.1 Pooled vertical/attitudinal AP measures by type

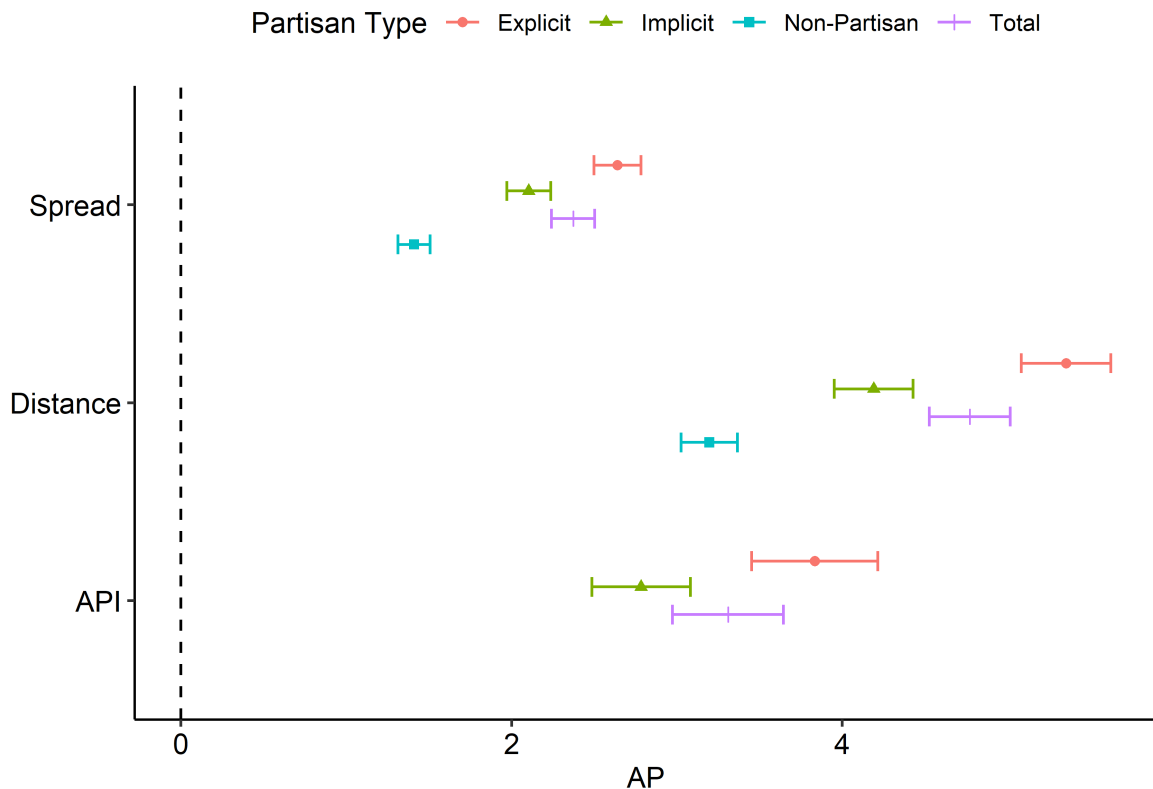


Figure 2: Attitudinal Affective Polarization in the EU25. This figure displays aggregated attitudinal AP scores across three operationalizations, categorized by partisan attachment type. API follows Reiljan (2020)'s and is calculated as the distance between in-group and average out-group feeling thermometer scores. Distance and Spread follow Wagner (2021), capturing the root mean square distance from the most-liked party and the overall standard deviation of party evaluations, respectively. All estimates are weighted by party vote shares within each country to reflect national party system compositions and then averaged across 25 European countries. [OLD: To ensure comparability across measures, thermometer scores (originally 0–100) are rescaled to a 0–10 range]. Explicit partisans are respondents reporting a psychological attachment to a party; Implicit partisans are those denying partisan attachment but reporting a vote choice; Total includes all individuals with a partisan anchor; and Non-Partisans are respondents without a reported party anchor. Error bars indicate 95% confidence intervals based on cross-national variance.

Source: [Code Notebook 3.2](#)

-> across measures, explicit partisans systematically higher AP than implicit

-> Non-partisans (which can be included in Wagner measures) systematically less polarized

-> restricting sample to explicit partisans (identifiers) yields higher aggregate AP

5.2 Country-level vertical/attitudinal AP measures by type

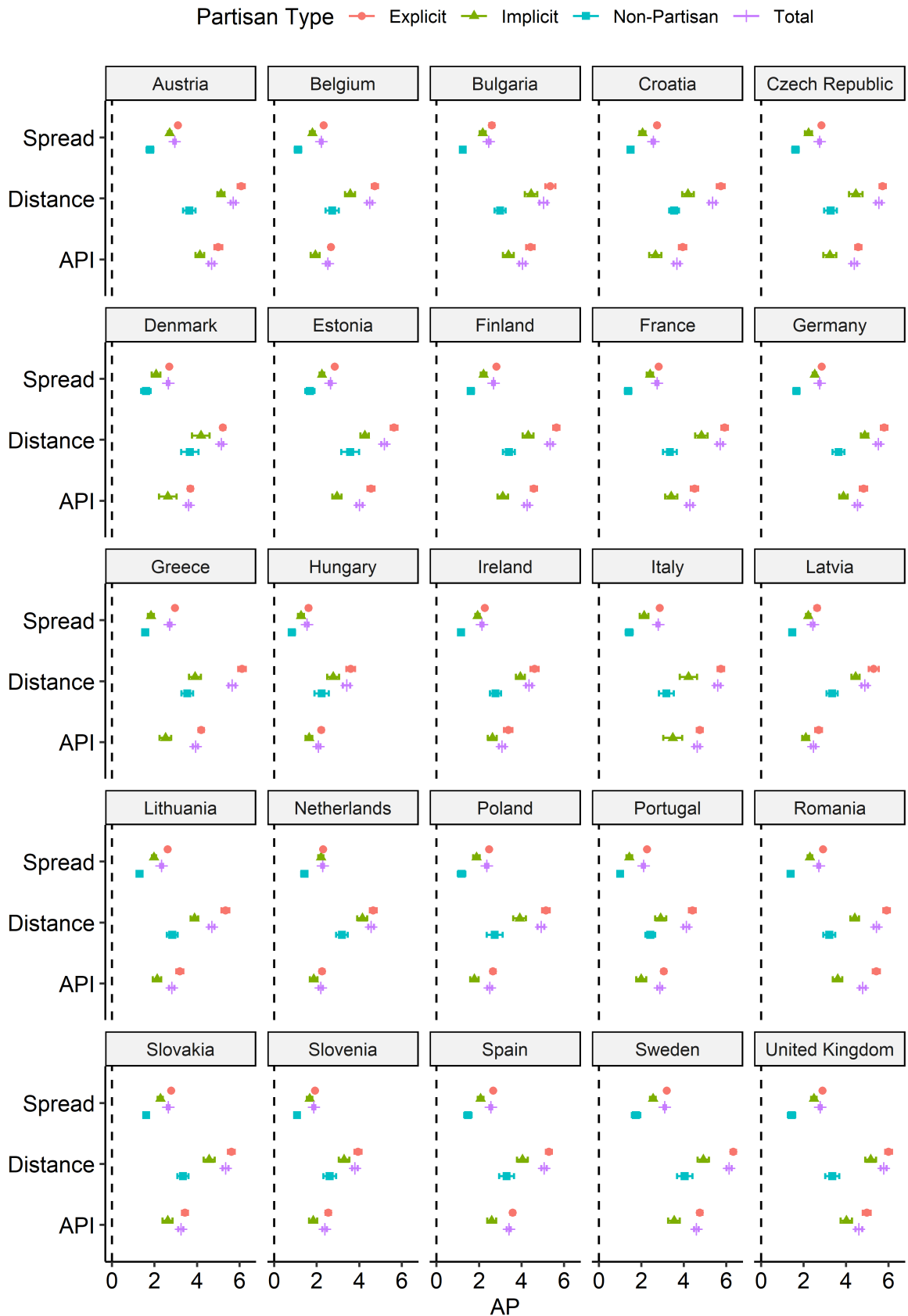


Figure 3: Attitudinal Affective Polarization across European Countries. This figure displays country-specific estimates of attitudinal affective polarization (AP) across three operationalizations and partisan attachment types. Attitudinal API follows Reiljan (2020) and is calculated as the distance between in-group and average out-group feeling thermometer scores. Attitudinal Distance and Spread follow wagner2021affective, capturing the root mean square distance from the most-liked party and the overall standard deviation of party evaluations, respectively. Estimates are calculated separately for each country and weighted by party vote shares to reflect national party system compositions. Thermometer scores (originally 0–100) are rescaled to a 0–10

Source: [Code Notebook 3.2](#)

- > pattern roughly consistent across countries (i.e., blue -> green -> orange)
- > cross-national variation in terms of magnitudes and within-country differences
- > With the varying country-specific distributions (Figure 1) in mind, AP estimates for all partisans (total) shift either close to that of explicit partisans (e.g. Denmark, Greece, Sweden) or stay centered between estimates for explicit and implicit partisans (e.g. Austria, Bulgaria, Ireland)

5.3 Pooled horizontal/behavioral AP measures by type

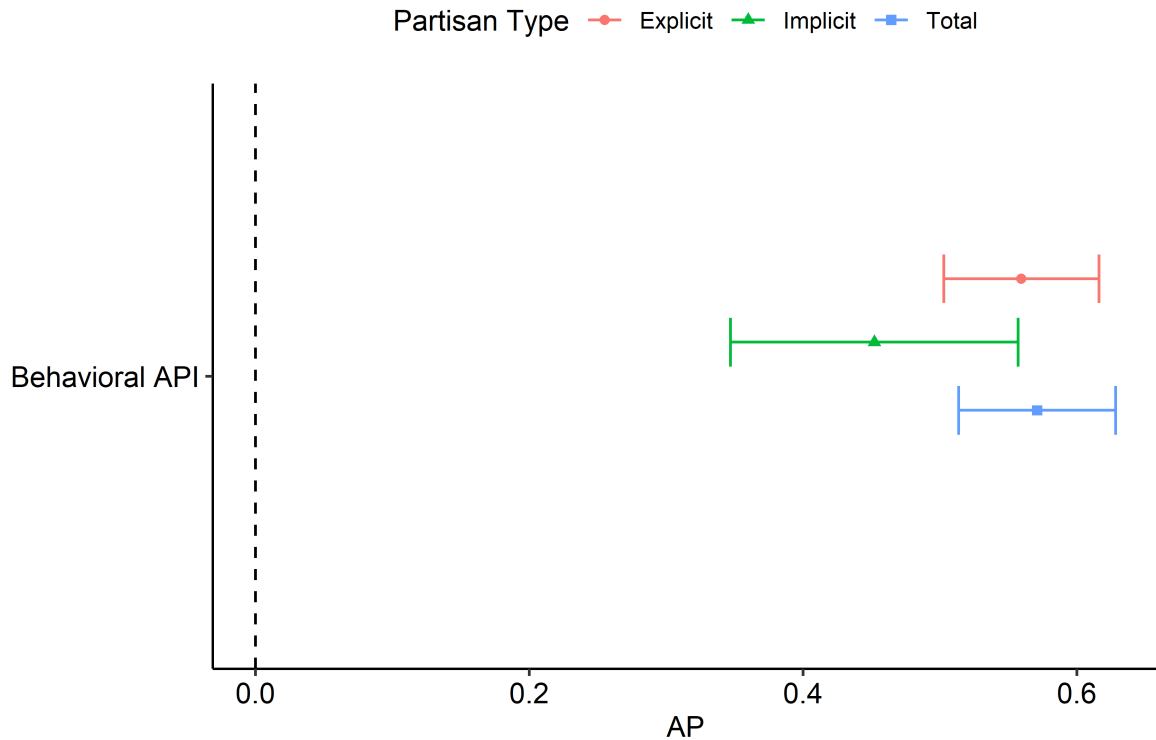


Figure 4: Behavioral Affective Polarization in the EU25. This figure displays aggregated behavioral affective polarization (AP) scores across partisan attachment types. Behavioral API (Token) measures the difference in expected token allocations between co-partisans and out-partisans in a conjoint allocation game. All estimates are weighted by party vote shares within each country to reflect national party system compositions and then averaged across 25 European countries. Explicit partisans are respondents reporting a psychological attachment to a party; Implicit partisans are those denying partisan attachment but reporting a vote choice; Total includes all individuals with a partisan anchor; and Non-Partisans are respondents without a reported party anchor. Error bars indicate 95% confidence intervals based on cross-national variance.

Source: [Code Notebook 3.2](#)

-> unlike in attitudinal AP, intervals for explicit and implicit overlap, hence no systematic difference

-> preliminary interpretation: according to data explicit attachment does not matter for behavioral AP

5.4 Country-level horizontal/behavioral AP measures by type

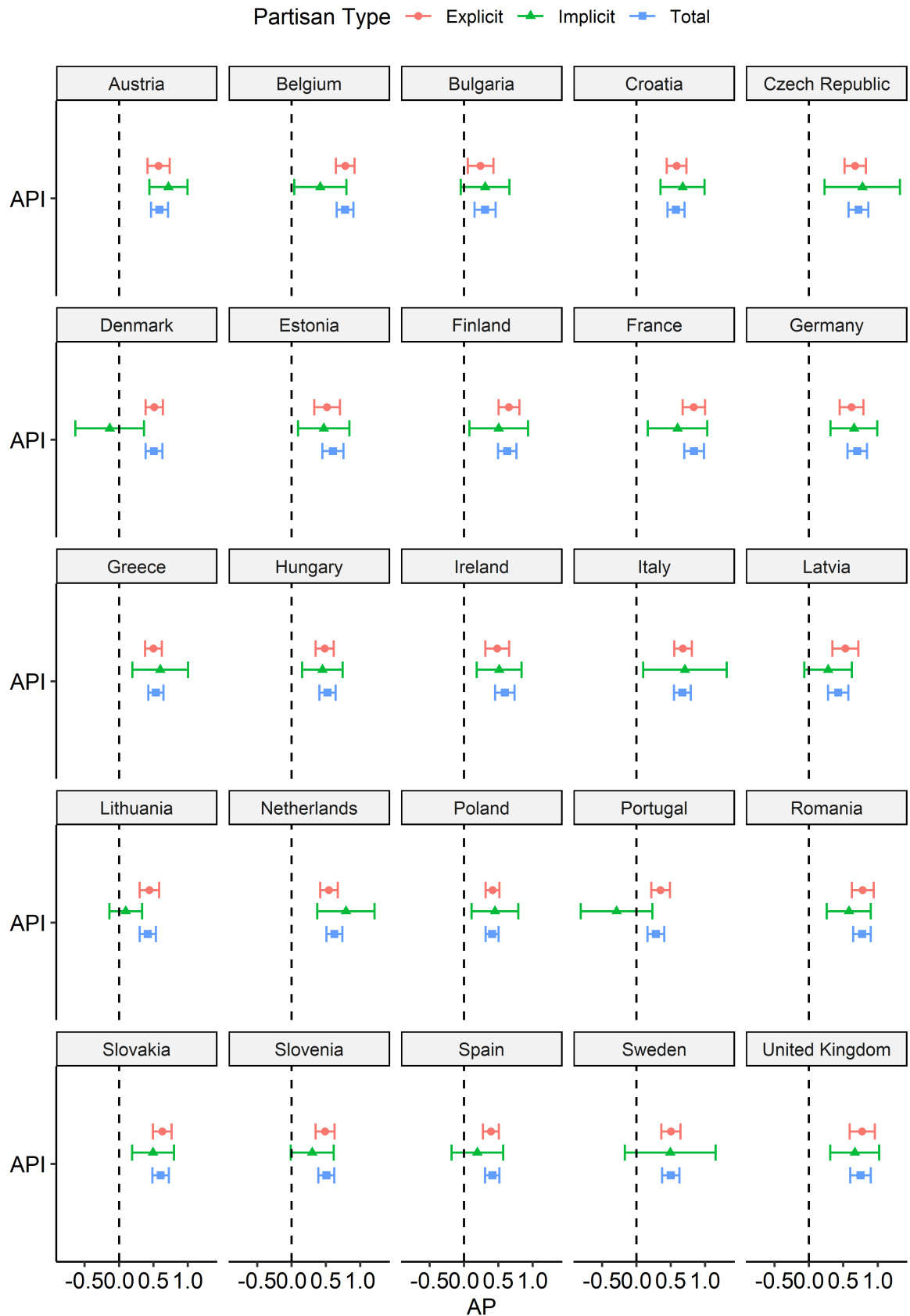


Figure 5: Behavioral Affective Polarization across European Countries. This figure displays country-specific estimates of behavioral affective polarization (AP) across partisan attachment types. Behavioral API (Token) measures the difference in expected token allocations between co-partisans and out-partisans in a conjoint allocation game. Estimates are calculated separately for each country and weighted by party vote shares to reflect national party system compositions. Explicit partisans are respondents reporting a psychological attachment to a party; Implicit partisans are those denying partisan attachment but reporting a vote choice; Total includes all individuals with a partisan anchor; and Non-Partisans are respondents without a reported party anchor.

- > pooled results generally consistent across countries
- > no systematic difference in behavioral AP between explicit and implicit
- > “strongest tendency” in Denmark, Lithuania and Portugal
- > otherwise almost no difference
- > estimate for total partisans drawn to explicit partisans (due to conjoint setup)

6 Limitations

This is so far mainly a collection of limitations for myself in order to not forget them in the end.

- Party-targeted attitudinal measures only

The attitudinal analysis relies exclusively on party-targeted thermometer ratings. It does not include direct citizen-targeted affect measures, limiting conclusions about horizontal AP at the attitudinal level.

- Assumption underlying party-size weighting

Weighting by electoral party size—common in the literature—implicitly assumes that the probability of encountering partisans is proportional to vote share. This may not hold if certain parties (e.g., radical challenger parties) are socially more or less visible than their electoral strength suggests.

- Cross-sectional design (2019)

The data capture a specific pre-pandemic political context and allow inference only about the level (state) of AP at that point in time. The design does not permit conclusions about temporal dynamics or trends.

- External validity of the experimental setting

Although the conjoint experiment approximates horizontal, citizen-level interactions, it remains an artificial setting. In real-world encounters, identities are rarely presented as explicit labels. Instead, they are inferred through cues and cognitive processes. This limits the direct generalizability of experimentally induced identity effects to everyday social interactions.

7 Conclusion

tbd

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9 Appendix

9.1 Sample descriptives

Table 1: Sample composition by country. Numbers denote respondents.

Country	N	Percent
Austria	1277	0.04
Belgium	1305	0.04
Bulgaria	982	0.03
Croatia	1240	0.04
Czech Republic	1135	0.04
Denmark	1200	0.04
Estonia	944	0.03
Finland	1160	0.04
France	1156	0.04
Germany	1188	0.04
Greece	1161	0.04
Hungary	986	0.03
Ireland	1061	0.04
Italy	1172	0.04
Latvia	1148	0.04
Lithuania	1265	0.04
Netherlands	1221	0.04
Poland	1198	0.04
Portugal	1187	0.04

Table 1: Sample composition by country. Numbers denote respondents.

Country	N	Percent
Romania	1480	0.05
Slovakia	1297	0.04
Slovenia	1135	0.04
Spain	1396	0.05
Sweden	1254	0.04
United Kingdom	1279	0.04
Total	29827	0.99

Source: [Code Notebook 3.1](#)

Table 2: Sample composition by country and gender. Numbers denote respondents.

Country	Male	Female	Other
Austria	645	630	2
Belgium	729	574	2
Bulgaria	463	518	1
Croatia	545	694	1
Czech Republic	515	618	2
Denmark	690	508	2
Estonia	331	611	2
Finland	586	567	7
France	562	594	0
Germany	587	597	4
Greece	588	572	1

Table 2: Sample composition by country and gender. Numbers denote respondents.

Country	Male	Female	Other
Hungary	493	492	1
Ireland	481	577	3
Italy	603	569	0
Latvia	415	733	0
Lithuania	462	803	0
Netherlands	642	577	2
Poland	540	658	0
Portugal	593	593	1
Romania	837	641	2
Slovakia	550	746	1
Slovenia	569	566	0
Spain	677	718	1
Sweden	648	602	4
United Kingdom	635	641	3
Total	14386	15399	42

Source: [Code Notebook 3.1](#)

Table 3: Sample composition by country and age group. Numbers denote respondents.

Country	18 to 25	26 to 35	36 to 45	46 to 55	56 to 65	66 to 75	> 75
Austria	144	177	249	263	286	142	0
Belgium	197	132	160	235	333	239	3
Bulgaria	57	195	231	250	212	33	0

Table 3: Sample composition by country and age group. Numbers denote respondents.

Country	18 to 25	26 to 35	36 to 45	46 to 55	56 to 65	66 to 75	> 75
Croatia	119	251	272	331	205	47	0
Czech Republic	88	187	227	219	281	126	0
Denmark	132	134	139	208	321	249	4
Estonia	56	168	164	283	265	7	0
Finland	125	178	210	242	259	137	1
France	123	176	242	267	279	51	0
Germany	125	184	200	240	313	117	0
Greece	77	202	376	318	136	34	0
Hungary	51	175	196	166	273	117	1
Ireland	128	216	223	187	169	120	0
Italy	104	196	264	195	294	98	2
Latvia	78	251	231	338	239	6	0
Lithuania	223	311	257	250	212	2	0
Netherlands	149	126	161	234	340	194	3
Poland	200	342	222	187	196	42	0
Portugal	112	260	289	231	218	67	1
Romania	128	339	373	347	202	65	0
Slovakia	139	223	282	290	250	108	1
Slovenia	97	186	246	284	234	72	0
Spain	125	293	355	305	228	66	1
Sweden	124	165	156	234	297	261	3

Table 3: Sample composition by country and age group. Numbers denote respondents.

Country	18 to 25	26 to 35	36 to 45	46 to 55	56 to 65	66 to 75	> 75
United Kingdom	121	184	196	234	268	250	1
Total	3022	5251	5921	6338	6310	2650	21

Source: [Code Notebook 3.1](#)

10 Schematic overview

11 Figures

Source: [Code Notebook 3.1](#)

11.1 Survey item wording (English version)

Table 4: Survey item wording (English version)

Variable	Question
Party	Do you consider yourself to be close to any particular political party?
Attachement	If so, which party do you feel close to?
Vote Intention	Which party will you give your 1st preference vote in this [2019 European] election?
Vote Choice	Which party did you give your 1st preference vote to in this [2019 European] election?

Variable	Question
Feeling Thermometer	We would like to get your feelings toward some political parties and political leaders in the news these days. Below there is a list of parties and people. I'd like you to rate them using something we call the feeling thermometer. Ratings between 50 degrees and 100 degrees mean that you feel favorable and warm toward the party or person. Ratings between 0 degrees and 50 degrees mean that you don't feel favorable toward the party or person and that you don't care too much for that party or person. You would rate the political party or the person at the 50 degree mark if you don't feel particularly warm or cold toward the person.

11.2 Experimental setup

Before the behavioral games, Hahm, Hilpert, and König (2024) presented respondents a short background information overview and instructions. For the dictator game, these were: *This game is played by pairs of individuals. Each pair is made up of a Player 1 and a Player 2. Each player will have some information about the other player, but you will not be told who the other players are during or after the experiment. The game is conducted as follows: A sum of 10 tokens will be provisionally allocated to Player 1 at the start of each round. Player 1 will then decide how much of the 10 tokens to offer to Player 2. Player 1 could give some, all, or none of the 10 tokens. Player 1 keeps all tokens not given to Player 2. Player 2 gets to keep all the tokens Player 1 offers. You will play this game three times with three different people.* In the trust game, the provided information and instruction were: *This game is played by pairs of individuals. Each pair is made up*

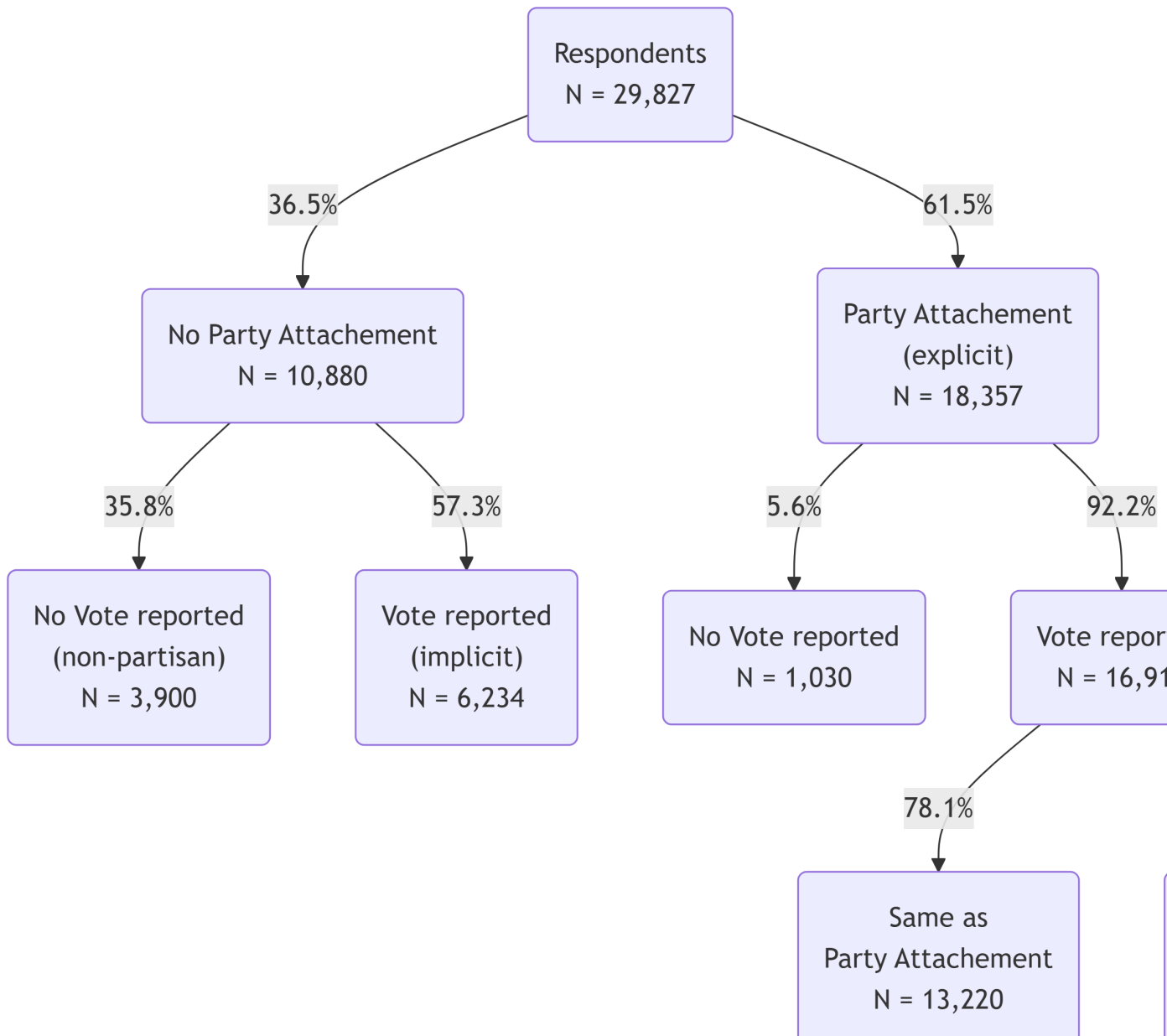


Figure 6: Distribution of respondents by reported party attachment and vote behavior. Percentages refer to the parent node.

of a Player 1 and a Player 2. Each player will have some information about the other player, but you will not be told who the other players are during or after the experiment. Each player will receive 10 tokens. Player 1 then has the opportunity to give a portion of his or her 10 tokens to Player 2. Player 1 could give some, all, or none of the 10 tokens. Whatever amount Player 1 decides to give to Player 2 will be tripled before it is passed on to Player 2. Player 2 then has the option of returning any portion of this tripled amount to Player 1. Then, the game is over. Player 1 receives whatever he or she keeps from the original 10 tokens, plus anything returned to him or her by Player 2. Player 2 receives their original 10 tokens, plus whatever he or she keeps after returning any portion of the tripled amount to Player 1. You will play this game three times, with three different people. The more tokens you obtain, the more successful you will be.

In both games respondents were shown a tabular overview of Player 2 after the instructions. Figure 7 shows an example of such a profile along with the interface respondents were provided to assign the 10 tokens. Each round, a new profile was displayed to respondents.

	Player 2
Nationality	United Kingdom
Age	18
Party Affiliation	Labour Party (Labour)
Gender	Female
Religion	Muslim
Social Class	Middle Class

So put the number of tokens you wish to keep in the box labeled "Player 1." Put the tokens you wish to go to Player 2 in the box labeled "Player 2."

Player 1 (You)	0	Token(s)
Player 2	0	Token(s)
Total	0	Token(s)

Figure 7: Example of potential co-player profile.

11.3 Distribution of Y

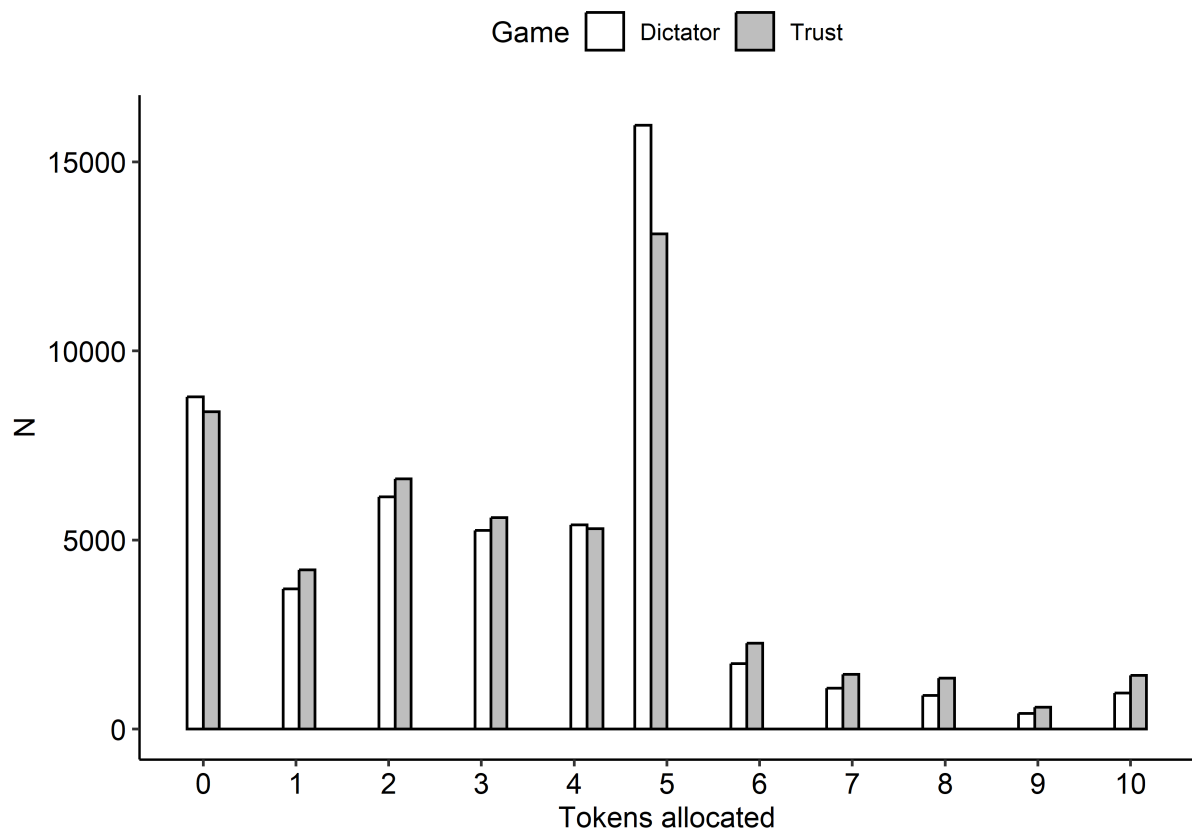


Figure 8: Distribution of token allocation by game. Dictator game: $Mean = 3.41$, $median = 4$, $SD = 2.35$. Trust game: $Mean = 3.48$, $median = 4$, $SD = 2.49$

Source: [Code Notebook 3.1](#)

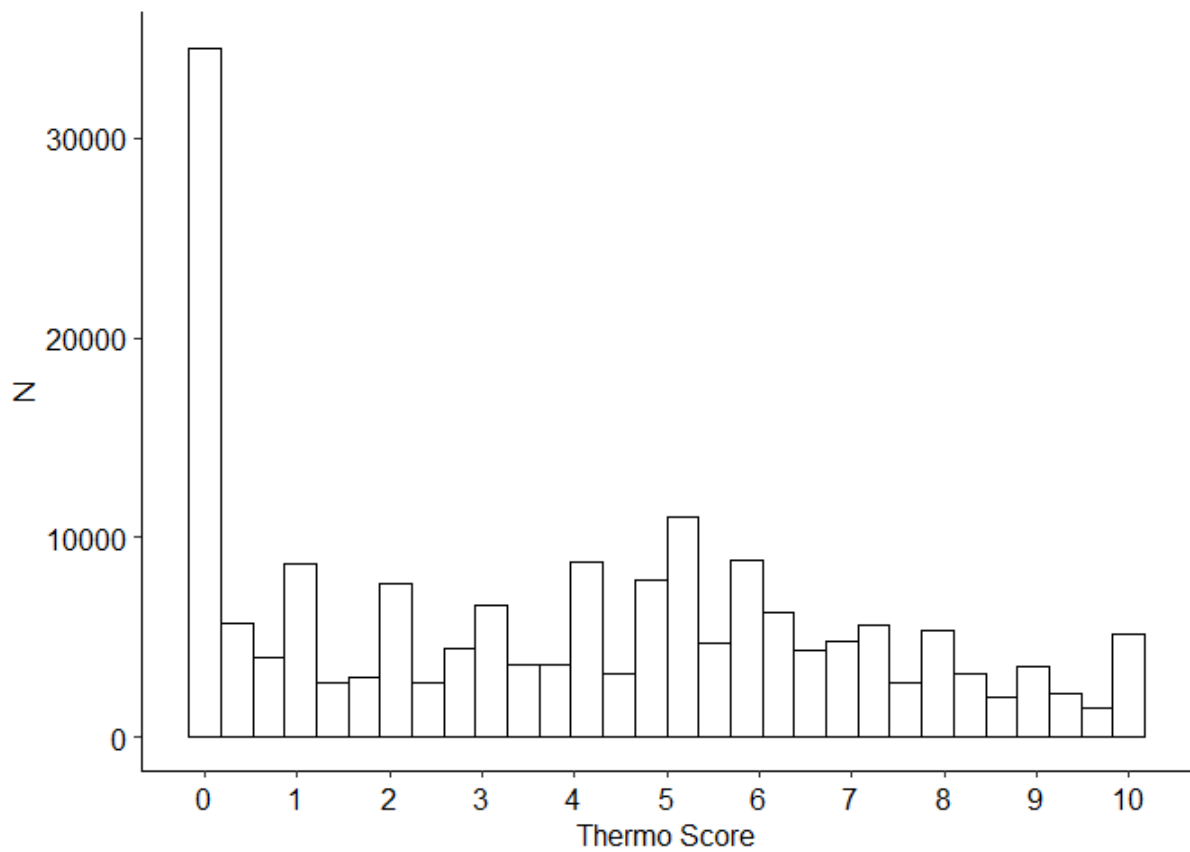


Figure 9: Distribution of thermometer ratings (rescaled from 0-100 to 0-10). $Mean = 3.89$, $median = 4$, $SD = 3.04$

Source: [Code Notebook 3.2](#)

11.4 Robustness